THE EXTERIOR PRODUCT OVER A FIELD IS ASSOCIATIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FC1 (sources/stacks-chow/pinned-original-source.tex line 14342).

Allowed standard prerequisites (editorial).

- Associativity of the exterior product over a field. Same chapter, Lemma labelled lemma-exterior-product-associative, cited at proof lines 14351-14353. Supplies the associativity of the untwisted exterior product.
- Commutation of the Gysin classes of the two partial diagonals. Same chapter, Lemma labelled lemma-gysin-commutes, cited at proof lines 14374-14381, with Remark labelled remark-restriction-bivariant for the restrictions. Gives Delta^! o Delta_12^! = Delta^! o Delta_23^!.
- The product defined by the diagonal class over a field. Same chapter, the construction preceding this lemma, invoked in the statement. Fixes the meaning of the product being shown associative.
- Commutation of the exterior product with the diagonal class. Same chapter, Remark labelled remark-commuting-exterior, cited at proof lines 14394-14399. Justifies the second and penultimate equalities.

Source reference. Original lemma 0FBZ: sources/stacks-chow/pinned-original-source.tex, source lines 14277–14277. Same-chapter pointer into the author environment lemma at source line 14277; invoked at proof line 14352 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0F9Z: sources/stacks-chow/pinned-original-source.tex, source lines 6184–6184. Same-chapter pointer into the author environment remark at source line 6184; invoked at proof line 14367 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBP: sources/stacks-chow/pinned-original-source.tex, source lines 12206–12206. Same-chapter pointer into the author environment lemma

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at source line 12206; invoked at proof line 14373 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FBY: sources/stacks-chow/pinned-original-source.tex, source lines 14261–14261. Same-chapter pointer into the author environment remark at source line 14261; invoked at proof line 14398 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *The product defined above is associative. More precisely, let k be a field, let X be smooth over k, let Y, Z, W be schemes locally of finite type over X, let $\alpha \in \mathrm{CH}_*(Y)$, $\beta \in \mathrm{CH}_*(Z)$, $\gamma \in \mathrm{CH}_*(W)$. Then $(\alpha \cdot \beta) \cdot \gamma = \alpha \cdot (\beta \cdot \gamma)$ in $\mathrm{CH}_*(Y \times_X Z \times_X W)$.*

Proof. By Lemma 0FBZ we have $(\alpha \times \beta) \times \gamma = \alpha \times (\beta \times \gamma)$ in $\mathrm{CH}_*(Y \times_k Z \times_k W)$. Consider the closed immersions

$$\Delta_{12} : X \times_k X \longrightarrow X \times_k X \times_k X, \quad (x, x') \mapsto (x, x, x')$$

and

$$\Delta_{23} : X \times_k X \longrightarrow X \times_k X \times_k X, \quad (x, x') \mapsto (x, x', x')$$

Denote $\Delta_{12}^!$ and $\Delta_{23}^!$ the corresponding bivariant classes; observe that $\Delta_{12}^!$ is the restriction (Remark 0F9Z) of $\Delta^!$ to $X \times_k X \times_k X$ by the map pr_{12} and that $\Delta_{23}^!$ is the restriction of $\Delta^!$ to $X \times_k X \times_k X$ by the map pr_{23}. Thus clearly the restriction of $\Delta_{12}^!$ by Δ_{23} is $\Delta^!$ and the restriction of $\Delta_{23}^!$ by Δ_{12} is $\Delta^!$ too. Thus by Lemma 0FBP we have

$$\Delta^! \circ \Delta_{12}^! = \Delta^! \circ \Delta_{23}^!$$

Now we can prove the lemma by the following sequence of equalities:

$$\begin{aligned}
(\alpha \cdot \beta) \cdot \gamma &= \Delta^!(\Delta^!(\alpha \times \beta) \times \gamma) \\
&= \Delta^!(\Delta_{12}^!((\alpha \times \beta) \times \gamma)) \\
&= \Delta^!(\Delta_{23}^!((\alpha \times \beta) \times \gamma)) \\
&= \Delta^!(\Delta_{23}^!(\alpha \times (\beta \times \gamma)) \\
&= \Delta^!(\alpha \times \Delta^!(\beta \times \gamma)) \\
&= \alpha \cdot (\beta \cdot \gamma)
\end{aligned}$$

All equalities are clear from the above except perhaps for the second and penultimate one. The equation $\Delta_{23}^!(\alpha \times (\beta \times \gamma)) = \alpha \times \Delta^!(\beta \times \gamma)$ holds by Remark 0FBY. Similarly for the second equation. □